1 Lyapunov stability theorem and Barbalat's lemma

Theorem 1. ***(Barbalat's lemma)***

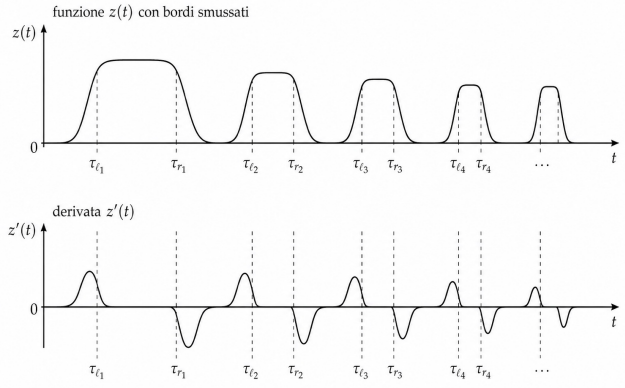
If a function $r(t) \in L_2 \cap L_\infty$ has a bounded derivative ($\dot{r}(t) \in L_\infty$), it converges asymptotically to zero.

Remark 2.

- The assumption $r \in L_\infty$ is stronger than necessary.
- The assumption $r(t) \in L_2 \bigcap L_\infty$ can be replaced by $r(t) \in L_p \bigcap L_\infty$, where $p \geq 1$.
- The bounded-derivative condition $\dot{r}(t) \in L_\infty$ matters. For example, consider a function consisting of rectangular pulses with areas $1, \frac{1}{2}, \frac{1}{4} \ldots$, spaced one second apart. The total integral is

$$\begin{aligned} \int_0^\infty f(t)\mathrm{dt} &= \sum_{k=0}^{\infty} \frac{1}{2^k} \\ &= \frac{1}{1-\frac{1}{2}} \\ &= 2 \end{aligned}$$

Hence $r \in L_1$. Since the signal is also bounded, we obtain $r \in L_1 \cap L_\infty, r \in L_2$.



Smooth the edges of the pulses while keeping their areas unchanged. As the function rises and falls over progressively shorter intervals, the magnitude of its derivative grows without bound. Thus, if $r(t) \in L_2 \bigcap L_\infty$ but $\dot{r}(t) \notin L_\infty$, the conclusion $\lim_{t\to\infty} r(t) = 0$ is not guaranteed.

2 Passive systems

Definition 3. *A square system is* ***passive*** *if it is dissipative with respect to the supply rate* $s(u, y) = u^T y$*, meaning that there exists a storage function* $S(x(t)) \geq 0$ *satisfying*

$$S(x(t_1)) \;\leq\; S(x(t_0)) + \int_{t_0}^{t_1} u^T(t) y(t) \cdot \mathrm{dt}$$

For $S \in C^1$*, the system is passive if*

$$S' \;\leq\; u^T y$$

Remark 4. In an electrical system with voltages as inputs and currents as outputs, the passivity supply rate represents total power. In a mechanical system with torques as inputs and moments as outputs, the supply rate represents mechanical power.

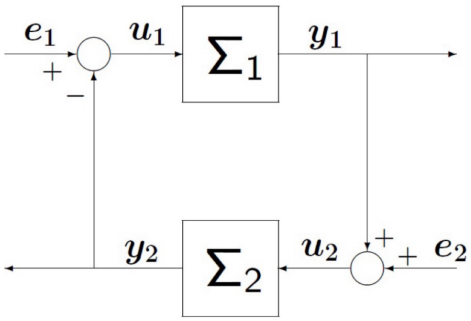
3 Passivity theorem

Theorem 5. *Consider the following nonlinear systems*

$$\Sigma_1 \quad \begin{aligned} \dot{x_1} &= f_1(x_1, u_1) \\ y_1 &= h_1(x_1, u_1) \end{aligned} \qquad \Sigma_2 \quad \begin{aligned} \dot{x_2} &= f_2(x_2, u_2) \\ y_2 &= h_2(x_2, u_2) \end{aligned}$$

with input signals u_1 and u_2 satisfying

$$u_1 = e_1 - y_2 \qquad u_2 = e_2 + y_1$$



1. *If Σ_1 and Σ_2 are passive with storage functions S_1 and S_2, then the interconnected system is passive with storage function $S = S_1 + S_2$.*

Proof.

The block diagram gives the input of the interconnected system as

$$e = \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}$$

the state as

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

and the output as

$$y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

By definition,

$$\begin{aligned} S_1(x_1(t_1)) &\leq S_1(x_1(t_0)) + \int_{t_0}^{t_1} u_1^T(\tau) y_1(\tau) \quad d\tau \\ S_2(x_2(t_1)) &\leq S_2(x_2(t_0)) + \int_{t_0}^{t_1} u_2^T(\tau) y_2(\tau) \quad d\tau \end{aligned}$$

or equivalently,

$$\begin{aligned} S_1(x_1(t_1)) &\leq S_1(x_1(t_0)) + \int_{t_0}^{t_1} (e_1(\tau) - y_2(\tau))^T y_1(\tau) \quad d\tau \\ S_2(x_2(t_1)) &\leq S_2(x_2(t_0)) + \int_{t_0}^{t_1} (e_2(\tau) + y_1(\tau))^T y_2(\tau) \quad d\tau \end{aligned}$$

Adding the two inequalities gives

$$\begin{aligned}
S_1(x_1(t_1)) + S_2(x_2(t_1)) &\leq S_1(x_1(t_0)) + \int_{t_0}^{t_1} (e_1(\tau) - y_2(\tau))^T y_1(\tau) \quad d\tau \\
&\quad + S_2(x_2(t_0)) + \int_{t_0}^{t_1} (e_2(\tau) + y_1(\tau))^T y_2(\tau) \quad d\tau \\
&= S_1(x_1(t_0)) + S_2(x_2(t_0)) + \int_{t_0}^{t_1} e_1^T(\tau) y_1(\tau) - y_2^T(\tau) y_1(\tau) \\
&\quad + e_2^T(\tau) y_2(\tau) + y_1^T(\tau) y_2(\tau) \quad d\tau \\
&= S_1(x_1(t_0)) + S_2(x_2(t_0)) + \int_{t_0}^{t_1} e_1^T(\tau) y_1(\tau) + e_2^T(\tau) y_2(\tau) \qquad d\tau \\
&= S_1(x_1(t_0)) + S_2(x_2(t_0)) + \int_{t_0}^{t_1} \begin{pmatrix} e_1^T(\tau) & e_2^T(\tau) \end{pmatrix} \begin{pmatrix} y_1(\tau) \\ y_2(\tau) \end{pmatrix} \qquad d\tau \\
&= S_1(x_1(t_0)) + S_2(x_2(t_0)) + \int_{t_0}^{t_1} e^T(\tau) y(\tau) \qquad d\tau
\end{aligned}$$

Setting $S(x) = S_1(x_1) + S_2(x_2)$ gives

$$S(x(t_1)) \leq S(x(t_0)) + \int_{t_0}^{t_1} e^T(\tau) y(\tau) \quad d\tau$$

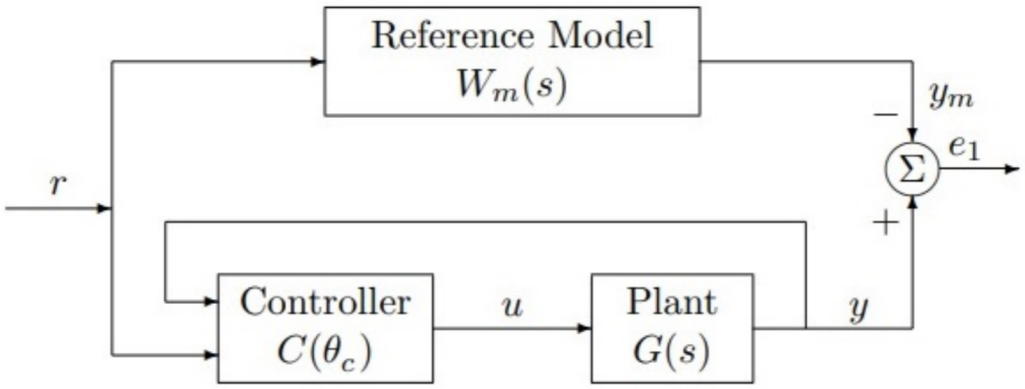
This proves the result.

□

4 MRAC

Model Reference Adaptive Control (MRAC) is one of the main methods used in adaptive control.

It consists of a **reference model** $W_m(s)$, a **parameterised controller** $C(\theta_c)$, and a **plant** $G(s)$. At each instant, the same reference r is applied to the reference model and the controller. The reference model produces the output y_m that the plant must track, while the controller generates the control input u applied to the plant.



MRAC is **direct** when an adaptation mechanism updates the controller parameters directly. It is **indirect** when estimates of the plant parameters are used to update the controller parameters.

Consider the scalar system

$$\dot{x} \;=\; a\,x + u$$

where x is measurable and a is an unknown constant.

If a were known, asymptotic stability could be achieved by setting

$$u \;=\; -a\,x - a_m x$$

where $a_m > 0$ is a constant chosen by the designer. Since a is unknown, there are two options.

- Estimate $k = a + a_m$ directly as $\hat{k}$, then set $u = -\hat{k}\,x$ (direct MRAC).
- Estimate the constant a directly, then set $u = -(\hat{a} + a_m)x$ (indirect MRAC).

The first method is analysed in Section 5 and the second in Section 6.

5 Direct MRAC

Consider the scalar system

$$\dot{x} \;=\; a\,x + u$$

with measurable state x and unknown constant parameter a. Direct MRAC estimates $k = a + a_m$ directly and then sets $u = -\hat{k}\,x$.

First, rewrite the dynamics of x in terms of k. Since

$$\begin{aligned} k &= a + a_m \\ a &= k - a_m \end{aligned}$$

substituting into the dynamics of the state x gives

$$\begin{aligned} \dot{x} &= a\,x + u \\ &= (k - a_m)x + u \\ &= -a_m\,x + k\,x + u \end{aligned}$$

The dynamics of the state estimate $\hat{x}$ are

$$\dot{\hat{x}} \;=\; -a_m\hat{x} + \hat{k}x + u$$

Define

$$u \;=\; -\hat{k}\,x$$

Then

$$\begin{aligned} \dot{x} &= -a_m\,x + k\,x + u \\ &= -a_m\,x + k\,x - \hat{k}x \\ &= -a_m\,x - \tilde{k}x \end{aligned}$$

where

$$\tilde{k} \;=\; \hat{k} - k$$

and

$$\begin{aligned} \dot{\hat{x}} &= -a_m\hat{x} + \hat{k}x + u \\ &= -a_m\hat{x} + \cancel{\hat{k}}x - \cancel{\hat{k}}x \\ &= -a_m\hat{x} \end{aligned}$$

It follows that

$$\hat{x}(t) \;=\; \hat{x}(0)e^{-a_m t}$$

For simplicity, if the initial state estimate is 0, we obtain

$$\hat{x}(t) \;=\; 0 \qquad \forall t \geq 0$$

and the state estimation error is

$$e(t) \;=\; \hat{x}(t) - x(t) = -x(t) \qquad \forall t \geq 0$$

The error dynamics are therefore

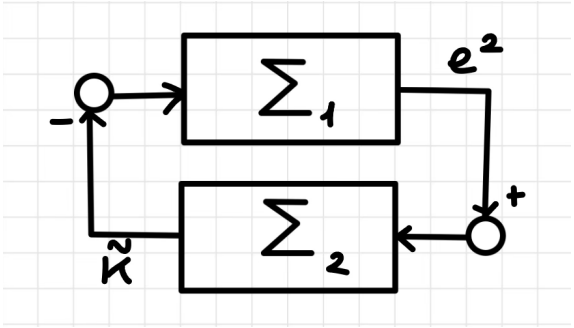
$$\begin{aligned} \dot{e} &= -\dot{x} \\ &= -(-a_m\, x - \tilde{k}x) \\ &= -(a_m + \tilde{k})(-x) \\ &= -(a_m + \tilde{k})e \end{aligned}$$

We now use passivity to make the error dynamics asymptotically stable. If $S_1 = \frac{1}{2}e^2$, then

$$\begin{aligned} \dot{S}_1 &= e\,\dot{e} \\ &= -(a_m + \tilde{k})e^2 \\ &= -a_m e^2 - \tilde{k}e^2 \end{aligned}$$

Since $a_m > 0$ by assumption, $-a_m e^2 \leq 0$, and the system is passive with input -$\tilde{k}$ and output e^2.

To stabilise the system, interconnect it with a system with input e^2 and output $\tilde{k}$ and cancel the term $-\tilde{k}e^2$.



Setting $S_2 = \frac{1}{2}\tilde{k}^2$ gives

$$\dot{S}_2 \;=\; \tilde{k}\dot{\tilde{k}}$$

To ensure $\dot{S}_2 = \tilde{k}e^2$, we require

$$\begin{aligned} \dot{\tilde{k}} &= e^2 \\ &= x^2 \end{aligned}$$

Recall that $\tilde{k} = \hat{k} - k$, with k constant. The update law for $\hat{k}$ must therefore be

$$\dot{\hat{k}} \;=\; x^2$$

Recall that

$$\dot{S}_{\text{tot}} \;=\; \dot{S}_1 + \dot{S}_2 = -a_m e^2 \leq 0$$

and that

$$S_{\text{tot}} \;=\; S_1 + S_2 = \frac{1}{2}e^2 + \frac{1}{2}\tilde{k}^2 \leq 0$$

The Lyapunov function is therefore bounded above, which gives $e, \tilde{k} \in L_\infty$. Since $e = -x$, we also obtain $x \in L_\infty$.

To prove that $e \in L_2$, note that

$$\begin{aligned} \dot{S}_{\text{tot}} &= -a_m e^2 \\ e^2 &= -\frac{\dot{S}_{\text{tot}}}{a_m} \end{aligned}$$

It follows that

$$\begin{aligned} \|e\|_2^2 &= \int_0^\infty e^2(t)\mathrm{dt} \\ &= \int_0^\infty -\frac{\dot{S}_{\text{tot}}}{a_m}\mathrm{dt} \\ &= -\lim_{t\to\infty} \frac{S_{\text{tot}}(t)}{a_m} + \frac{S_{\text{tot}}(0)}{a_m} \\ &\leq \frac{S_{\text{tot}}(0)}{a_m} \\ &< \infty \end{aligned}$$

since $S(t) \geq 0 \quad \forall t \geq 0$ and $a_m > 0$, and $S_{\text{tot}}(0)$ is fixed.

Finally, since

$$\dot{e} = -(a_m + \tilde{k})e$$

and $\tilde{k}, e \in L_\infty$, we obtain $\dot{e} \in L_\infty$.

By Barbalat's lemma,

$$\begin{aligned} \lim_{t\to\infty} x(t) &= -\lim_{t\to\infty} e(t) \\ &= 0 \end{aligned}$$

In summary,

Theorem 6. *consider the scalar system*

$$\dot{x} = a\,x + u$$

where a is an unknown constant.

If the control law is

$$u = -\hat{k}\,x$$

and the adaptation law is

$$\dot{\hat{k}} = x^2$$

then the closed-loop system is asymptotically stable.

6 Indirect MRAC

Consider the scalar system

$$\dot{x} \;=\; a\,x + u$$

with measurable state x and unknown constant parameter a, and let

$$u \;=\; -(\hat{a} + a_m)x$$

We seek an update law for $\hat{a}$ that makes $x_e = 0$ asymptotically stable.

The I&I method first defines the error dynamics.

$$z \;=\; \hat{a} - a$$

It follows that

$$\begin{aligned}
\dot{x} &= a\,x + u \\
&= a\,x - (\hat{a} + a_m)x \\
&= -a_m x - (\hat{a} - a)x \\
&= -a_m x - z\,x
\end{aligned}$$

Since $\hat{a}$ can depend only on known variables, introduce the dynamic variable ξ and set

$$\hat{a} \;=\; \beta(x, \xi)$$

The error dynamics become

$$\begin{aligned}
\dot{z} &= \dot{\hat{a}} - \dot{a} \\
&= \frac{d\beta(x,\xi)}{dx}\dot{x} + \frac{d\beta(x,\xi)}{d\xi}\dot{\xi} \\
&= \frac{d\beta(x,\xi)}{dx}(-a_m x - z\,x) + \frac{d\beta(x,\xi)}{d\xi}\dot{\xi} \\
&= -\frac{d\beta(x,\xi)}{dx}a_m x - \frac{d\beta(x,\xi)}{dx}z\,x + \frac{d\beta(x,\xi)}{d\xi}\dot{\xi}
\end{aligned}$$

For the error dynamics $\dot{z}$ to depend only on z, impose

$$-\frac{d\beta(x,\xi)}{dx}a_m x + \frac{d\beta(x,\xi)}{d\xi}\dot{\xi} \;=\; 0$$

It follows that

$$\dot{z} \;=\; -\frac{d\beta(x,\xi)}{dx}x\,z$$

Setting $V = \frac{1}{2}z^2$ gives

$$\begin{aligned}
\dot{V} &= z\,\dot{z} \\
&= -\frac{d\beta(x,\xi)}{dx}x\,z^2
\end{aligned}$$

To ensure $z \in L_\infty$, V must be bounded above, which requires $\dot{V} \leq 0$. It follows that $\frac{d\beta(x,\xi)}{dx} x \geq 0$.

- Setting

$$\begin{aligned} \frac{d\beta(x,\xi)}{dx} x &= x^2 \\ \frac{d\beta(x,\xi)}{dx} &= x \\ \beta(x,\xi) &= \frac{1}{2}x^2 + f(\xi) \end{aligned}$$

gives

$$\dot{z} = -x^2 z$$

For the equation

$$-a_m \frac{d\beta(x,\xi)}{dx} x + \frac{d\beta(x,\xi)}{d\xi}\dot{\xi} = 0$$

to hold, we require

$$-a_m x^2 + \frac{df(\xi)}{d\xi}\dot{\xi} = 0$$

For simplicity, choose $f(\xi) = \xi$.

$$\begin{aligned} -a_m x^2 + \dot{\xi} &= 0 \\ \dot{\xi} &= a_m x^2 \end{aligned}$$

- Alternatively, if we set

$$\begin{aligned} \frac{d\beta(x,\xi)}{dx} x &= \frac{x^2}{1+x^2} \\ \frac{d\beta(x,\xi)}{dx} &= \frac{x}{1+x^2} \\ \beta(x,\xi) &= \frac{1}{2}\log(1+x^2) + f(\xi) \end{aligned}$$

we obtain

$$\dot{z} = -\frac{x^2}{1+x^2} z$$

For the equation

$$-a_m \frac{d\beta(x,\xi)}{dx} x + \frac{d\beta(x,\xi)}{d\xi}\dot{\xi} = 0$$

to hold, we require

$$-a_m \frac{x^2}{1+x^2} + \frac{df(\xi)}{d\xi}\dot{\xi} = 0$$

For simplicity, choose $f(\xi) = \xi$.

$$\begin{aligned} -a_m \frac{x^2}{1+x^2} + \dot{\xi} &= 0 \\ \dot{\xi} &= a_m \frac{x^2}{1+x^2} \end{aligned}$$

We now show that the system

$$\dot{x} = -a_m x - z\,x$$

is asymptotically stable. First, we prove that $z\,x \in L_2$. There are two cases.

- If $\dot{\hat{a}} = \frac{1}{2}x^2 + \xi \quad \Rightarrow \quad \dot{z} = -x^2 z$, set $V = \frac{1}{2}z^2$.

$$\begin{aligned} \dot{V} &= z\,\dot{z} \\ &= -x^2 z^2 \end{aligned}$$

Since $V > 0$ and $\dot{V} \leq 0$, V is bounded, which gives $z \in L_\infty$. We also have

$$x^2 z^2 = -\dot{V}$$

Integrating from 0 to ∞ gives

$$\begin{aligned} \|x\,z\|_2^2 &= \int_0^\infty -\dot{V}\mathrm{dt} \\ &= -V(\infty) + V(0) \\ &\leq V(0) \\ &< \infty \end{aligned}$$

since $V > 0$ and $V(0)$ is fixed.

- If $\dot{\hat{a}} = \frac{1}{2}\log(1+x^2) + \xi \quad \Rightarrow \quad \dot{z} = -\frac{x^2}{1+x^2} z$, set $V = \frac{1}{2}\log(1+x^2) - z$.

$$\begin{aligned} \dot{V} &= \frac{x}{1+x^2}\dot{x} - \dot{z} \\ &= \frac{x}{1+x^2}(-a_m x - z\,x) - \left(-\frac{x^2}{1+x^2} z\right) \\ &= -a_m \frac{x^2}{1+x^2} - \frac{\cancel{x^2}}{1+\cancel{x^2}} z + \frac{\cancel{x^2}}{1+\cancel{x^2}} z \\ &= -a_m \frac{x^2}{1+x^2} \\ &\leq 0 \end{aligned}$$

It follows that V is bounded, so $\log(1+x^2)$ is bounded. Since the logarithm is increasing, $x \in L_\infty$.

Setting $V = \frac{1}{2}z^2$ gives

$$\begin{aligned} \dot{V} &= z\,\dot{z} \\ &= -\frac{x^2 z^2}{1+x^2} \end{aligned}$$

It follows that

$$\begin{aligned}\int_0^\infty \frac{x^2z^2}{1+x^2}\mathrm{dt} &= \int_0^\infty -\dot{V}\mathrm{dt} \\ &= -V(\infty)+V(0) \\ &\leq V(0) \\ &< \infty\end{aligned}$$

since $V(t)\geq 0 \forall t\geq 0$ and $V(0)$ is fixed.

Since

$$\begin{aligned}\int_0^\infty x^2z^2\mathrm{dt} &\leq \alpha\int_0^\infty \frac{x^2z^2}{1+x^2}\mathrm{dt} \\ &\Updownarrow \\ x^2z^2 &\leq \alpha\frac{x^2z^2}{1+x^2} \\ 1 &\leq \frac{\alpha}{1+x^2} \\ 1+x^2 &\leq \alpha\end{aligned}$$

and because $x\in L_\infty$, we obtain $u=z\,x\in L_2$.

Given $u\in L_2$, we obtain $x\in L_2\cap L_\infty$.

Recall that

$$\dot{x} = -a_m\,x+u$$

so

$$x(t) = e^{-a_m t}x_0+\int_0^t e^{-a_m(t-\tau)}u(\tau)d\tau$$

where $e^{-a_m t}x_0\to 0$ since $a_m>0$. Also, for two functions $f,g\in C^1$, the Cauchy-Schwarz inequality gives

$$\left|\int_a^b f(t)g(t)\mathrm{dt}\right| \leq \int_a^b f^2(t)\mathrm{dt}\int_a^b g^2(t)\mathrm{dt}$$

and hence

$$\begin{aligned}\int_0^t |e^{-a_m(t-\tau)}u(\tau)|d\tau &\leq \sqrt{\int_0^t e^{-2a_m(t-\tau)}d\tau\int_0^t u^2(\tau)d\tau} \\ &= \sqrt{\frac{e^{-2a_m(t-\tau)}}{2a_m}\Big|_0^t\int_0^t u^2(\tau)d\tau} \\ &= \sqrt{\frac{1-e^{-2a_m t}}{2a_m}\int_0^t u^2(\tau)d\tau} \\ &\to \sqrt{\frac{1}{2a_m}\int_0^t u^2(\tau)d\tau} \\ &< \infty\end{aligned}$$

since $u \in L_\infty$. We therefore obtain $x \in L_\infty$.

Setting $V = \frac{1}{2}x^2$ gives

$$\begin{aligned} \dot{V} &= x\,\dot{x} \\ &= x(-a_m x + u) \\ &= -a_m x^2 + u\,x \end{aligned}$$

Since

$$(-a_m x + u)^2 = a_m^2 x^2 + u^2 - 2a_m x\,u \geq 0$$

we obtain

$$\begin{aligned} a_m^2 x^2 + u^2 - 2a_m x\,u &\geq 0 \\ \frac{2a_m x\,u}{2a_m} &\leq \frac{a_m^2 x^2 + u^2}{2a_m} \\ x\,u &\leq \frac{\cancel{a_m^2} x^2}{2\cancel{a_m}} + \frac{u^2}{2a_m} \\ &= \frac{a_m x^2}{2} + \frac{u^2}{2a_m} \end{aligned}$$

It follows that

$$\begin{aligned} \dot{V} &= -a_m x^2 + u\,x \\ &\leq -a_m x^2 + \frac{a_m x^2}{2} + \frac{u^2}{2a_m} \\ &\leq -\frac{a_m x^2}{2} + \frac{u^2}{2a_m} \\ \frac{a_m x^2}{2} &\leq -\dot{V} + \frac{u^2}{2a_m} \\ x^2 &\leq -\frac{2}{a_m}\dot{V} + \frac{\cancel{2}}{a_m}\frac{u^2}{\cancel{2}a_m} \\ &= -\frac{2}{a_m}\dot{V} + \frac{u^2}{a_m^2} \end{aligned}$$

Integrating from 0 to ∞ gives

$$\begin{aligned} \int_0^\infty x^2 \mathrm{dt} &\leq \int_0^\infty \left(-\frac{2}{a_m}\dot{V} + \frac{u^2}{a_m^2} \right) \mathrm{dt} \\ &= -\frac{2}{a_m}\int_0^\infty \dot{V}\mathrm{dt} + \frac{1}{a_m^2}\int_0^\infty u^2 \mathrm{dt} \\ &= -\frac{2}{a_m}(V(\infty) - V(0)) + \frac{1}{a_m^2}\int_0^\infty u^2 \mathrm{dt} \\ &= -\frac{2}{a_m}V(\infty) + \frac{2}{a_m}V(0) + \frac{1}{a_m^2}\int_0^\infty u^2 \mathrm{dt} \\ &\leq \frac{2}{a_m}V(0) + \frac{1}{a_m^2}\int_0^\infty u^2 \mathrm{dt} \\ &< \infty \end{aligned}$$

since $V \geq 0, u \in L_2$ and $V(0)$ is fixed.

Finally, since $x, z \in L_\infty$, we have $\dot{x} = -a_m x + x\,z \in L_\infty$. By Barbalat's lemma, $z \to 0$, or equivalently, $\hat{a} \to a$.

In summary,

Theorem 7. *consider the scalar system*

$$\dot{x} \;=\; a\,x + u$$

with measurable state x and unknown constant a. If

$$u \;=\; -(a_m + \hat{a})x$$

where $a_m > 0$ is chosen by the designer and

$$\begin{aligned} \hat{a} \;&=\; \frac{1}{2}x^2 + \xi \\ \dot{\xi} \;&=\; a_m\,x^2 \end{aligned}$$

or

$$\begin{aligned} \hat{a} \;&=\; \frac{1}{2}\log(1+x^2) + \xi \\ \dot{\xi} \;&=\; a_m \frac{x^2}{1+x^2} \end{aligned}$$

then

$$\lim_{t\to\infty} \hat{a}(t) \;=\; a$$

7 Assignment

Consider the system

$$\dot{x} \;=\; a\,x + u$$

with measurable state x and unknown constant parameter a.

Assuming $a = 1$ and $a_m = 1$, design the MRAC and I&I adaptive controllers discussed above. Compare the two closed-loop systems in terms of response speed, control input magnitude, and behaviour under measurement noise $d(t) = 0.1\sin\left(\frac{t}{5}\right)$.

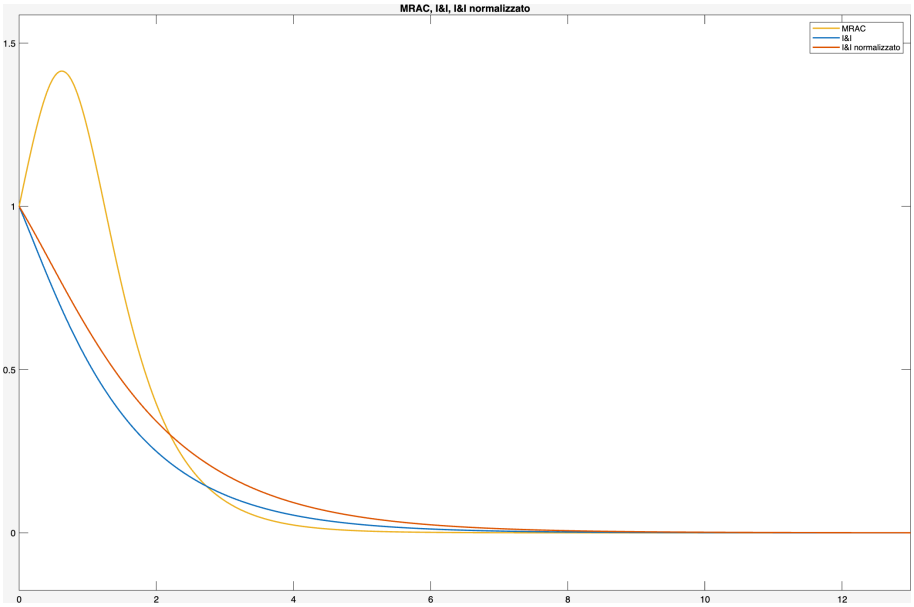
Finally, compare the two controllers when

$$a = 1 + \frac{1}{10}\sin(10t) \qquad \vee \qquad a = 1 + 10\sin\left(\frac{t}{10}\right)$$

8 $a = 1$ without disturbances

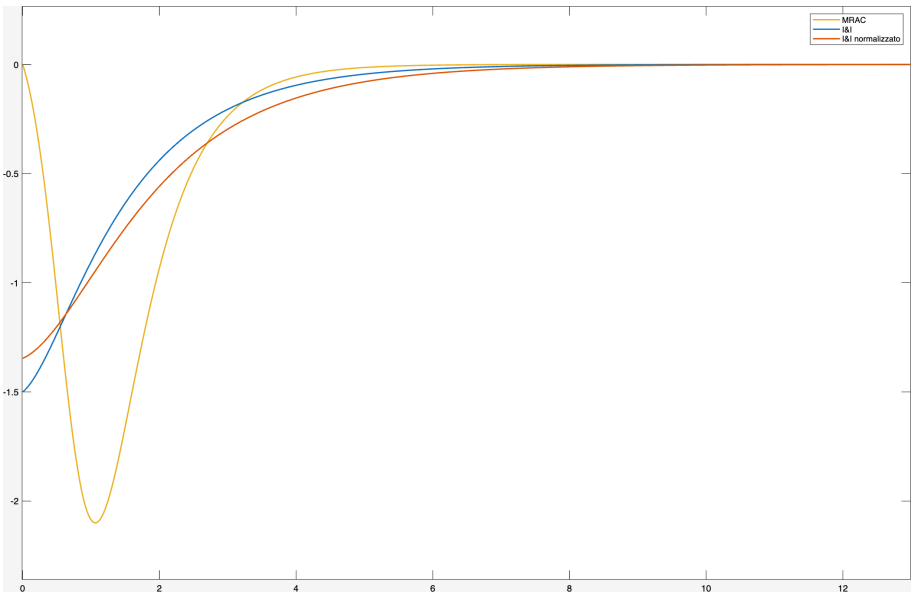
For $a = 1$ in the absence of disturbances, the results are as follows.

- Under MRAC, the state x converges asymptotically to zero in about $6s$ after an overshoot of 1.4.
- Under I&I, the state x converges asymptotically to zero in about $8s$ without overshoot.
- Under normalised I&I, the state x converges asymptotically to zero in about $11s$ without overshoot.



The peak control inputs are as follows.

- The MRAC control input u has a peak magnitude of 2.1.
- The I&I control input u has a peak magnitude of 1.5.
- The normalised I&I control input u has a peak magnitude of 1.35.



9 $a = 1$ with disturbance $d(t) = 0.1 \sin\left(\frac{t}{5}\right)$

If the state measurement is affected by the disturbance $d(t) = 0.1 \sin\left(\frac{t}{5}\right)$, the state and control input approach a steady-state response that follows the disturbance $d(t)$, with persistent oscillations around zero.

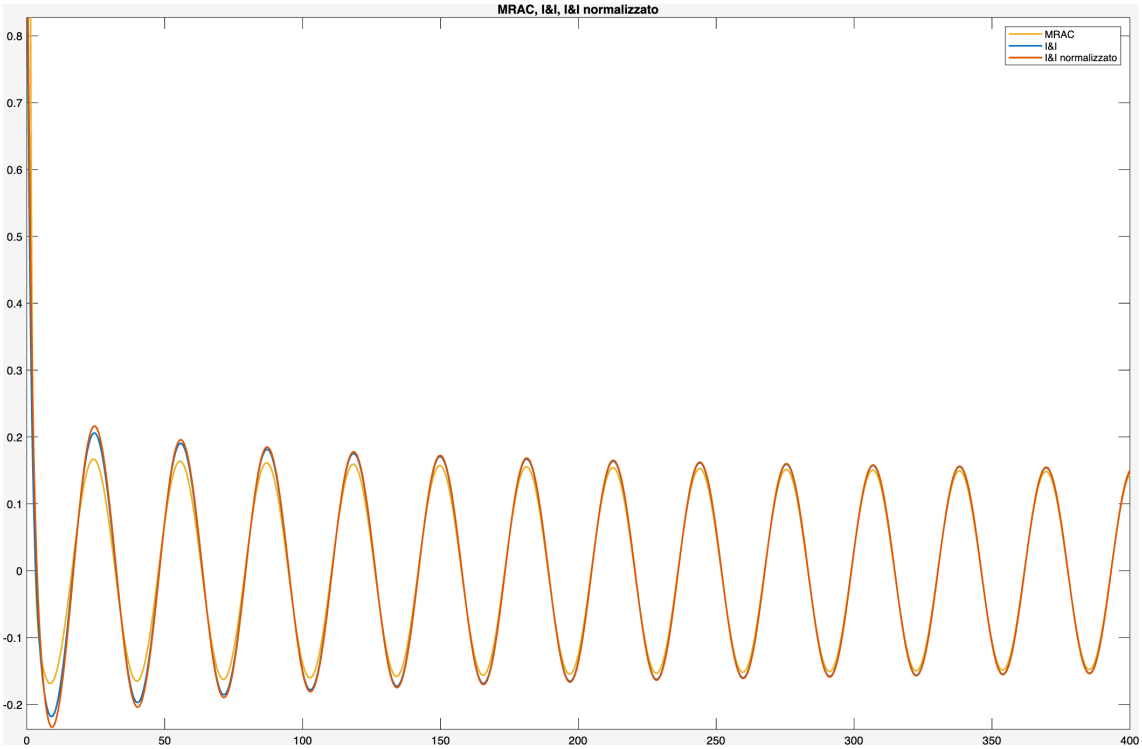


Figure 1. State x

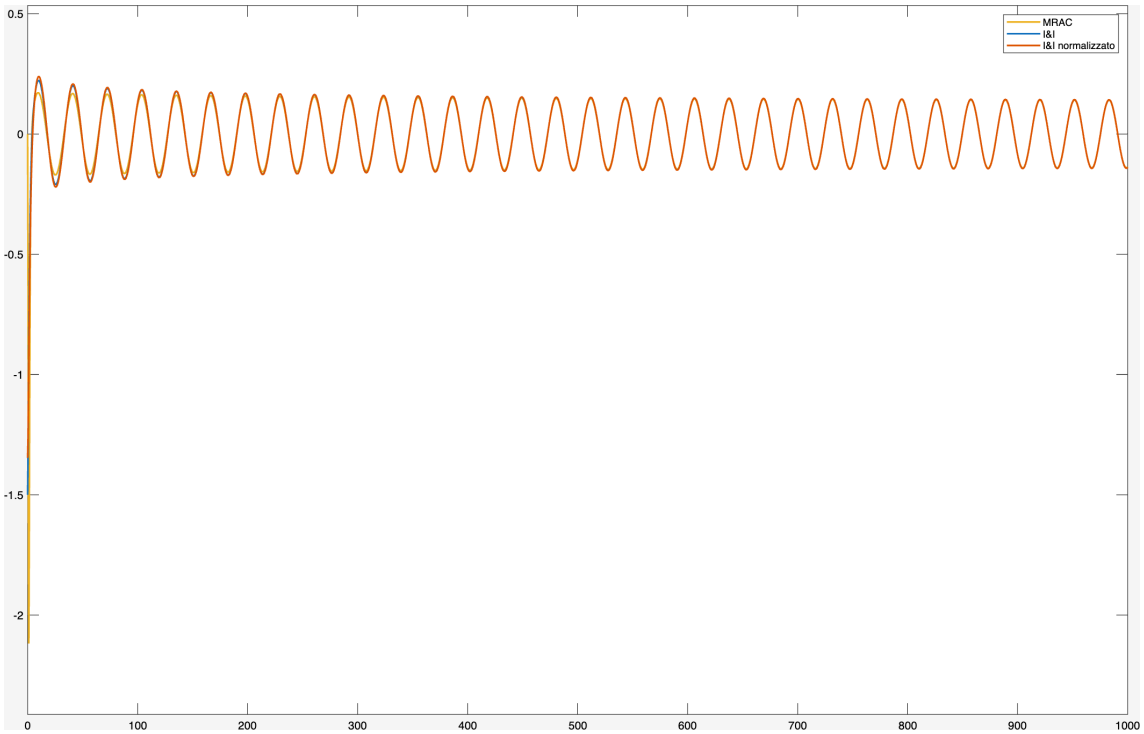
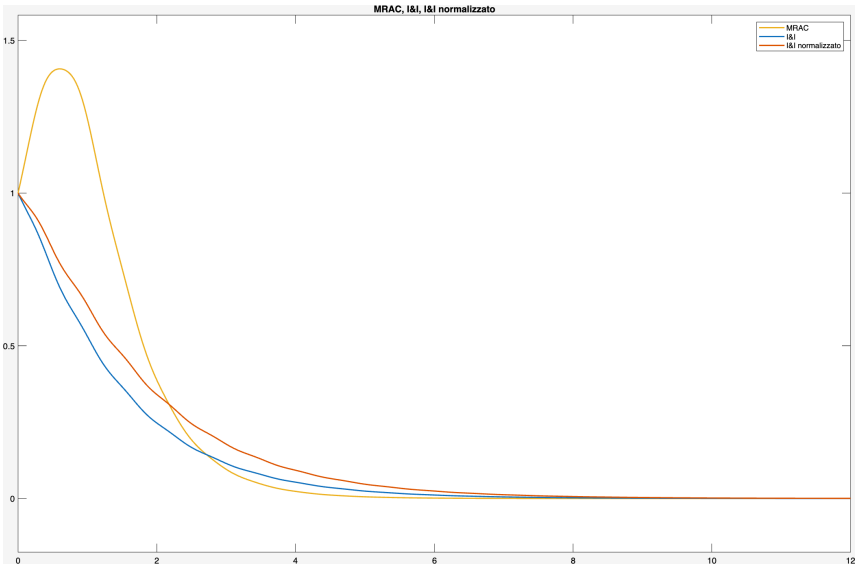


Figure 2. Control input u

10 $a = 1 + \frac{1}{10}\sin(10t)$ without disturbances

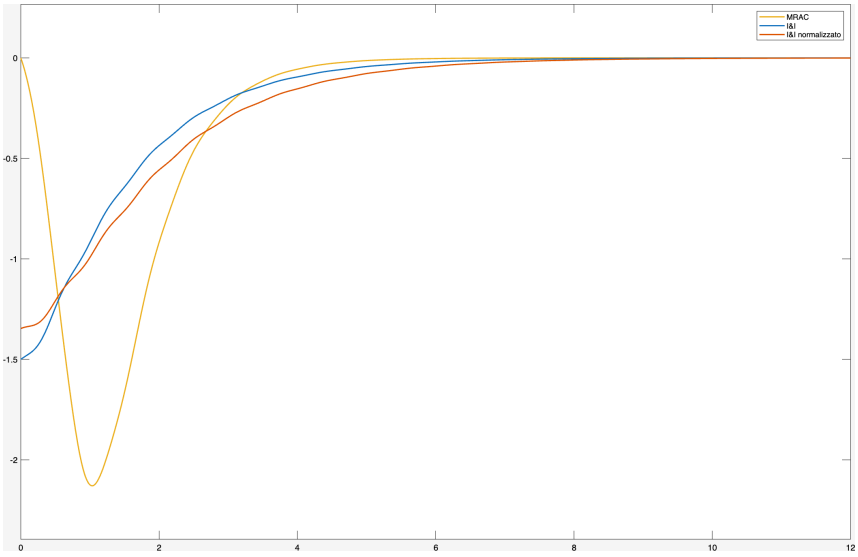
For $a = 1 + \frac{1}{10}\sin(10t)$ in the absence of disturbances, the results are as follows.

- Under MRAC, the state x converges asymptotically to zero in about $6s$ after an overshoot of 1.4.
- Under I&I, the state x converges asymptotically to zero in about $8s$ without overshoot and shows a slight change in slope during the transient.
- Under I&I, the state x converges asymptotically to zero in about $11s$ without overshoot and shows a slight change in slope during the transient.



The peak control inputs are as follows.

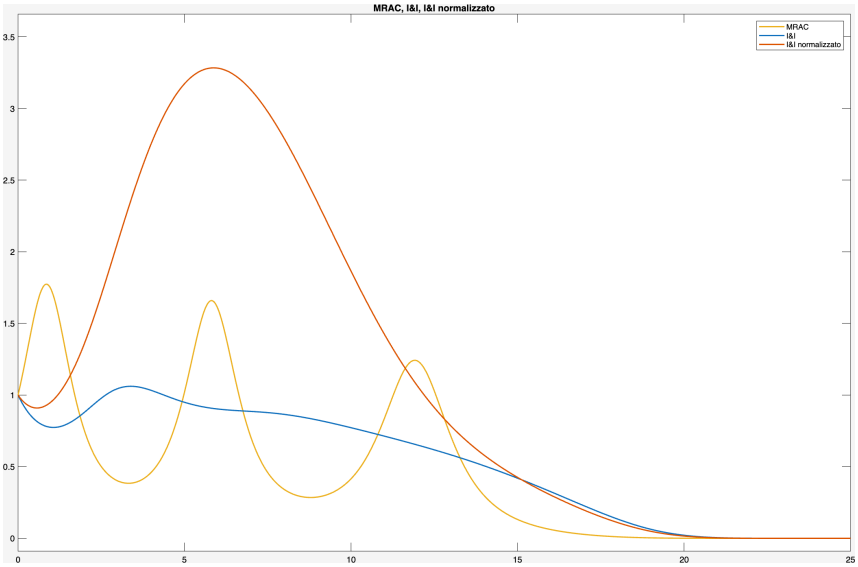
- The MRAC control input u has a peak magnitude of 2.1.
- The I&I control input u has a peak magnitude of 1.5 and shows a slight change in slope during the transient.
- The normalised I&I control input u has a peak magnitude of 1.35 and shows a slight change in slope during the transient.



11 $a = 1 + 10\sin\left(\frac{1}{10}t\right)$ without disturbances

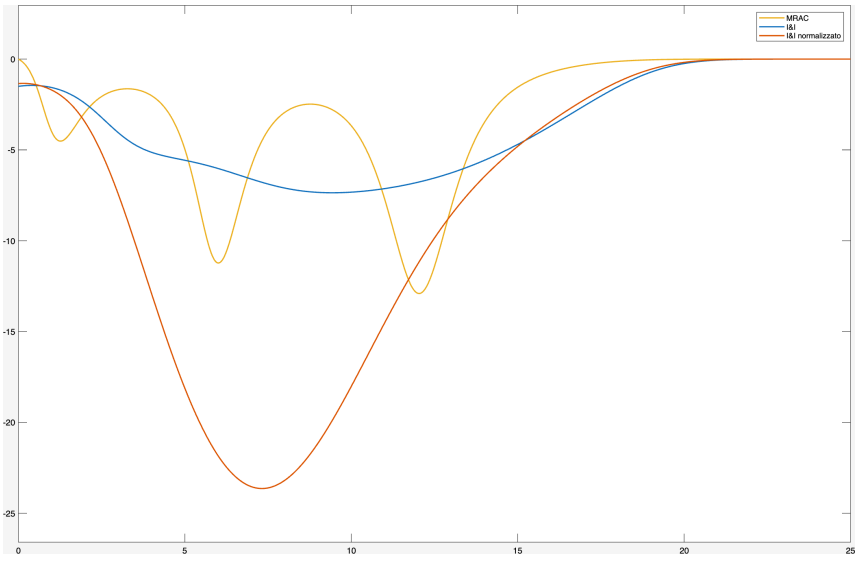
For $a = 1 + 10\sin\left(\frac{1}{10}t\right)$ in the absence of disturbances, the results are as follows.

- Under MRAC, the state x converges asymptotically to zero in about $20s$ after an overshoot of 1.8.
- Under I&I, the state x converges asymptotically to zero in about $22s$, with an overshoot of 1.6.
- Under I&I, the state x converges asymptotically to zero in about $22s$, with a maximum overshoot of 3.3.



The peak control inputs are as follows.

- The MRAC control input u has a peak magnitude of 12.9.
- The I&I control input u has a peak magnitude of 7.36 and shows some irregular variation.
- The normalised I&I control input u has a peak magnitude of 23.64.



12 Conclusions

The simulations yielded the following observations.

- With no disturbances and a constant parameter ($a = 1$), all three approaches stabilise the system asymptotically. MRAC converges fastest, but produces significant state overshoot and requires the greatest control effort. I&I and normalised I&I converge without overshoot, with slightly longer settling times and less control effort.

- With an additive disturbance in the state measurement, all three approaches produce a sinusoidal steady-state response in both the state and the control input. The response follows the disturbance, showing that the persistent measurement error propagates into the control input.

- When the parameter varies rapidly with time,

$$a(t) = 1 + \frac{1}{10}\sin(10t),$$

  performance is broadly similar to the constant-parameter case. Convergence times and peak control inputs are nearly the same, while the I&I approaches show only slight irregularities during the transient. In these simulations, all three algorithms tolerate high-frequency, small-amplitude parameter variations.

- When the parameter varies slowly and with a large amplitude,

$$a(t) = 1 + 10\sin\left(\frac{t}{10}\right),$$

  performance deteriorates for all controllers. Convergence takes up to about $20 - 22$ s, overshoot becomes more pronounced, and the required control effort increases considerably. MRAC produces the largest overshoot and the highest peak control input. I&I uses the lowest control effort, although its transient response is slightly less smooth.

Overall, the simulations show the following trade-offs.

- MRAC offers the fastest convergence, with greater overshoot and a larger control input.

- Standard I&I offers the best balance of convergence speed, low or zero overshoot, and control effort.

- Normalised I&I stabilises the system, but is generally slower and can require considerably more control effort than the other approaches when the parameter varies substantially.